5. Reactions on a surface

- Reaction mechanisms on metal surfaces involve
  - adsorption
  - chemical reaction at the surface
  - desorption
- In this section of the course, we shall assume mass transfer to reach the surface is rapid.
- We wish to derive the form of the rate law.
  - In Sections 5.1-5.4 we shall assume the chemical reaction is the slow step;
  - In Section 5.5, we will consider the possibility that adsorption or desorption is slow.

5.1 Langmuir-Hinshelwood mechanism

- The Langmuir-Hinshelwood mechanism assumes:
  1. Chemical reaction takes place between adsorbed species on the surface.
  2. The surface reaction is the rate determining step.
  3. Adsorption isotherms obey the assumptions of the Langmuir model.
- For simplicity consider the overall (irreversible) reaction

  $A_{(g)} + B_{(g)} \longrightarrow AB_{(g)}$
- The steps in the mechanism (using * to denote a surface site) are

Step 1	$A_{(g)} + * \rightleftharpoons A_{(*)}$	FAST
Step 2	$B_{(g)} + * \rightleftharpoons B_{(*)}$	FAST
Step 3	$A_{(*)} + B_{(*)} \rightarrow AB_{(*)} + *$	SLOW
Step 4	$AB_{(*)} \rightleftharpoons AB_{(g)} + *$	FAST

- The overall rate of reaction will simply be the rate of slow step 3, so

$$r = k_s \theta_A \theta_B$$

- Because step 1 is assumed to be fast, it will be at pseudo-equilibrium and so θ_A is given by an adsorption isotherm.
  - Similarly θ_B and θ_{AB} are given by adsorption isotherms.
- The Langmuir adsorption isotherm for mixtures (with competitive adsorption on the same sites) gives

$$\theta_A = \frac{K_A P_A}{(1 + K_A P_A + K_B P_B + K_{AB} P_{AB})}$$

$$\theta_B = \frac{K_B P_B}{(1 + K_A P_A + K_B P_B + K_{AB} P_{AB})}$$

- Hence the Langmuir-Hinshelwood mechanism predicts in this case the following rate law

$$r = k_s \theta_A \theta_B = \frac{k_s K_A K_B P_A P_B}{(1 + K_A P_A + K_B P_B + K_{AB} P_{AB})^2}$$

- Many reactions on surfaces follow the Langmuir-Hinshelwood mechanism reasonably well.
  - We shall examine the implication of this form of rate law in Sections 5.3 and 5.4.

5.2 Eley-Rideal mechanism

- The Eley-Rideal mechanism assumes:
  1. Chemical reaction takes place between a gas phase species and an adsorbed species.
  2. This reaction is the rate determining step.
  3. Adsorption isotherms obey the assumptions of the Langmuir model.
- For simplicity consider the overall (irreversible) reaction

  $A_{(g)} + B_{(g)} \longrightarrow AB_{(g)}$
- If B does not adsorb on the surface, then the steps in the mechanism are

Step 1	$A_{(g)} + * \rightleftharpoons A_{(*)}$	FAST
Step 2	$A_{(*)} + B_{(g)} \rightarrow AB_{(*)}$	SLOW
Step 3	$AB_{(*)} \rightleftharpoons AB_{(g)} + *$	FAST

- The overall rate of reaction will simply be the rate of slow step 2, so

  $$r = k\theta_A P_B$$
- Because step 1 is assumed to be fast, it will be at pseudo-equilibrium and so θ_A is given by an adsorption isotherm

  $$\theta_A = \frac{K_A P_A}{(1 + K_A P_A + \cancel{K_B P_B} + K_{AB} P_{AB})}$$
- Hence the Eley-Rideal mechanism predicts in this case the following rate law

  $$r = k\theta_A P_B = \frac{k K_A P_A P_B}{(1 + K_A P_A + K_{AB} P_{AB})}$$
- Very few reactions follow this mechanism, but it is still widely discussed in chemical engineering and catalysis literature.

5.3 Overall order of reaction

- For a single-step reaction, the kinetic order of reaction with respect to species A is the value of n_A in the equation

  $$r = k P_A^{n_A} P_B^{n_B} \ldots$$
- For a multi-step reaction, an overall kinetic order may be defined in a similar manner. The kinetic order of reaction with respect to reagent A is the value of n_A in the equation

  $$r = k_{overall} P_A^{n_A} P_B^{n_B} \ldots$$
- This may equivalently be written

  $$\ln r = \ln k_{overall} + n_A \ln P_A + n_B \ln P_B \ldots$$

- Values of n_A can be found from the gradient of a graph of $\ln r$ against $\ln P_A$, with other parameters being kept constant.
- Mathematically, this is equivalent to saying

$$n_A = \left(\frac{\partial \ln r}{\partial \ln P_A}\right)_{T,P_B\ldots} = P_A\left(\frac{\partial \ln r}{\partial P_A}\right)_{T,P_B\ldots}$$

Example

- Consider the Langmuir-Hinshelwood mechanism for $A_{(g)} + B_{(g)} \longrightarrow AB_{(g)}$. We showed earlier that in this case

$$r = k_s\theta_A\theta_B = \frac{k_s K_A K_B P_A P_B}{(1 + K_A P_A + K_B P_B + K_{AB} P_{AB})^2}$$

- What does this equation predict?

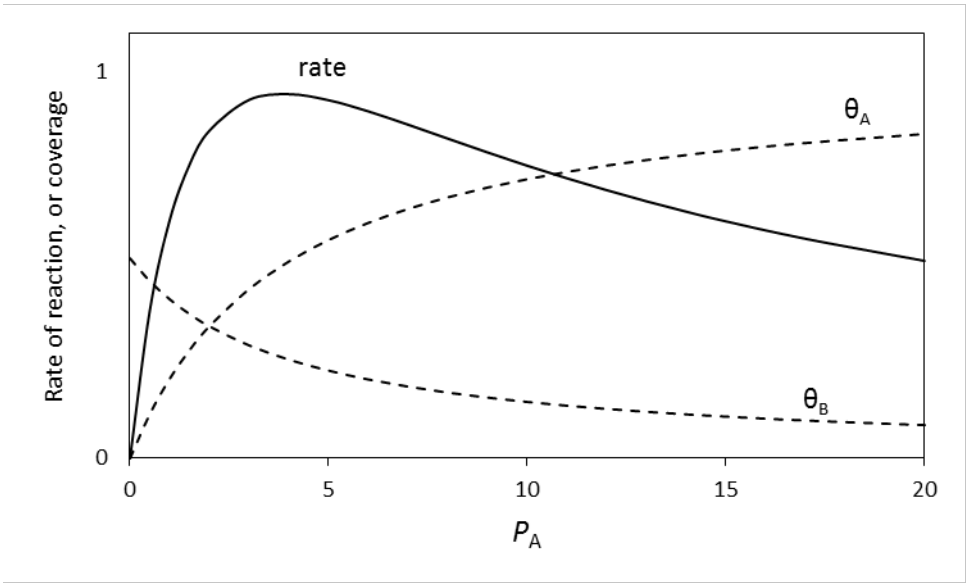


Figure: Rate of reaction while varying P_A (at fixed P_B and fixed T)

- At low P_A, there is little coverage of the surface with A, and so reaction is slow.
- At high P_A, the surface is mainly covered with A and there is little B on the surface, and so the reaction may be slow.
- The kinetic order of reaction with respect to A is

$$n_A = P_A\left(\frac{\partial \ln r}{\partial P_A}\right)_{T,P_B\ldots}$$

- In this special case, this gives (exercise for reader!):

$$n_A = 1 - 2\theta_A$$

  - $n_A \to 1$ as $\theta_A \to 0$, and so the reaction may be 1st order in A

- $n_A \to -1$ as $\theta_A \to 1$, and so the reaction may be –1st order with respect to A
- In general, n_A lies in the range $-1 < n_A < +1$; it will not normally be an integer.

- At some point, there will be an optimum rate of reaction.
  - The maximum corresponds to $dr/dP_A = 0$, or equivalently is when $n_A = 0$.

5.4 Overall activation energy

- For a single-step reaction, the activation energy of a reaction is defined through the Arrhenius equation for the rate constant

$$k = A_0 \exp\left(\frac{-E_{act}}{RT}\right)$$

- For a multi-step reaction, an overall activation energy may be defined in a similar manner

$$r = k_{overall} P_A^{n_A} P_B^{n_B} \ldots$$

with

$$k_{overall} = A_0 \exp\left(\frac{-E_{act}}{RT}\right)$$

- This may equivalently be written

$$\ln r = \ln A_0 - \frac{E_{act}}{R}\left(\frac{1}{T}\right) + n_A \ln P_A + n_B \ln P_B \ldots$$

- Values of E_{act} can be found from the gradient of a graph of $\ln r$ against $1/T$ (with all partial pressures being kept constant).
- Mathematically, this is equivalent to saying the overall activation energy obeys:

$$E_{act} = -R\left(\frac{\partial \ln r}{\partial(1/T)}\right)_{P_A,P_B\ldots} = RT^2\left(\frac{\partial \ln r}{\partial T}\right)_{P_A,P_B\ldots}$$

- For reactions on surfaces, values of E_{act} obtained in this way may be a combination of the activation energies for the individual steps in the reaction mechanism:
  - Note that the surface reaction rate constant k_s is expected to obey

$$k_s = A_s \exp\left(\frac{-E_s}{RT}\right)$$

  - Note that the adsorption equilibrium constants will obey the van't Hoff equation

$$\frac{d \ln K_A}{dT} = \frac{\Delta H_{ads,A}}{RT^2}$$

Example

- Consider the Langmuir-Hinshelwood mechanism for $A_{(g)} + B_{(g)} \longrightarrow AB_{(g)}$. We showed earlier that in this case

$$r = k_s \theta_A \theta_B = \frac{k_s K_A K_B P_A P_B}{(1 + K_A P_A + K_B P_B + K_{AB} P_{AB})^2}$$

- The overall activation energy of the reaction is

$$E_{\text{act}} = RT^2 \left(\frac{\partial \ln r}{\partial T}\right)_{P_{\text{A}},P_{\text{B}}\ldots}$$

- In this special case, this gives (exercise for reader!)

$$E_{\text{act}} = E_{\text{s}} + (1 - 2\theta_{\text{A}})\Delta H_{\text{ads,A}} + (1 - 2\theta_{\text{B}})\Delta H_{\text{ads,B}} - 2\theta_{\text{AB}}\Delta H_{\text{ads,AB}}$$

  - Remembers that enthalpies of adsorption are always negative.
- At low T, the activation energy will be a large positive number.
  - For instance, E_{act} may be greater than E_{s} if θ_{A} or θ_{B} are high
- At high T, the activation energy will be smaller:
  - For instance $E_{\text{act}} \to E_{\text{s}} + \Delta H_{\text{ads,A}} + \Delta H_{\text{ads,B}}$ as $\theta_{\text{A}} \to 0$, $\theta_{\text{B}} \to 0$, $\theta_{\text{AB}} \to 0$
- Depending on the magnitudes of the values:
  - E_{act} may become negative at high temperatures (low coverages). Physically this corresponds to the rate of reaction decreasing when the temperature goes up, because too few species are adsorbed on the surface for efficient reaction.

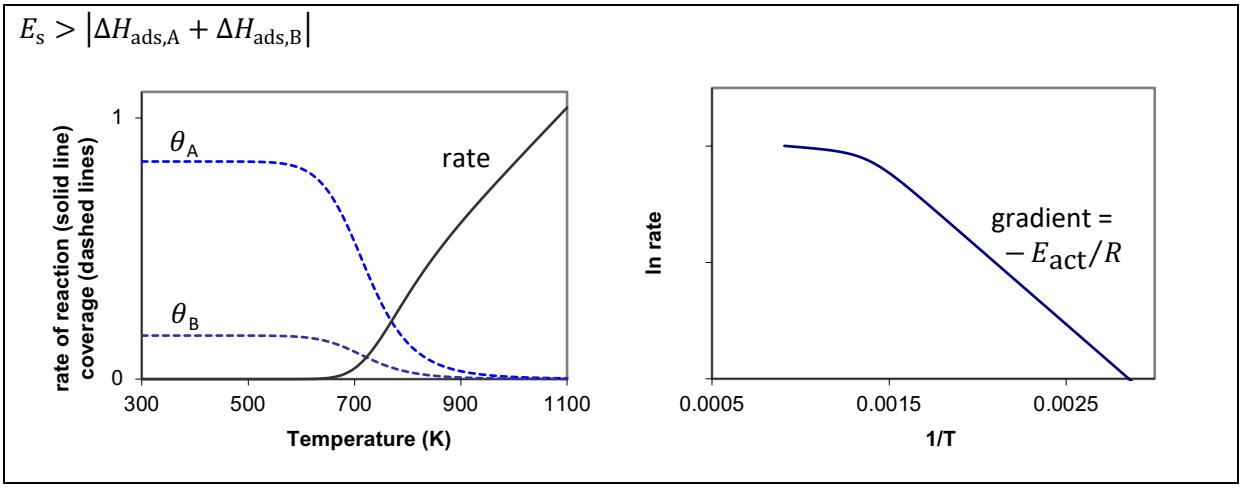


Figure: Example of how rate of reaction varies with temperature for one case of Langmuir-Hinshelwood kinetics, $E_{\text{s}} > |\Delta H_{\text{ads,A}} + \Delta H_{\text{ads,B}}|$

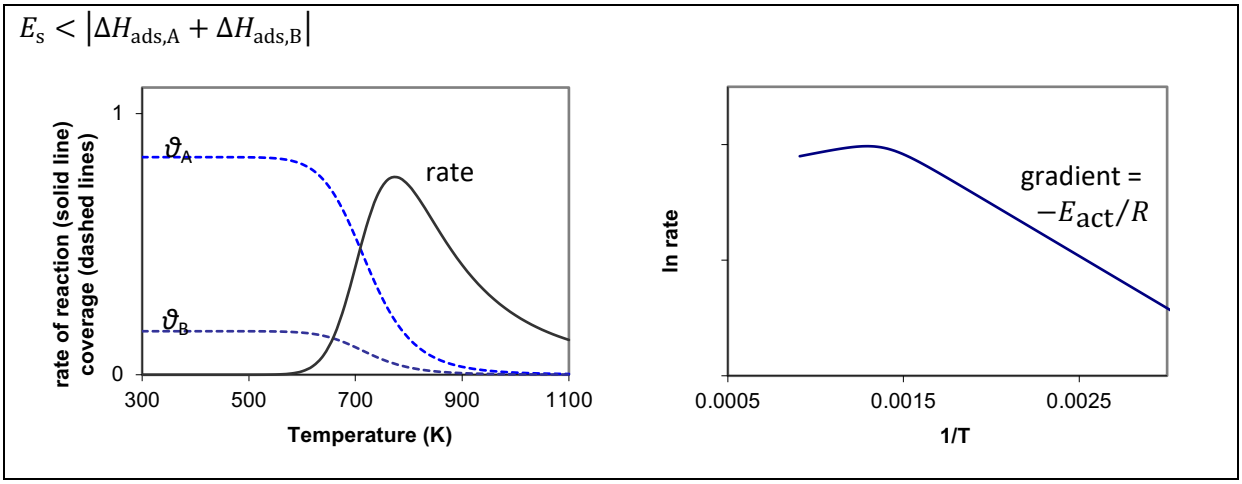
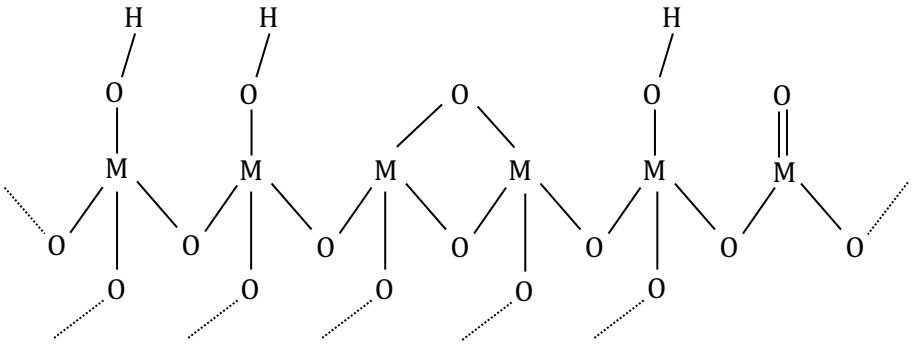


Figure: Example of how rate of reaction varies with temperature for one case of Langmuir-Hinshelwood kinetics, $E_s < |\Delta H_{ads,A} + \Delta H_{ads,B}|$

5.5 Mechanism for reaction on metal oxides

- Reaction mechanisms on metal oxides (and metal sulfides) tend to be more complicated than those on metal surfaces.
- Metal oxide surfaces are usually complex – each oxygen atom requires two bonds to it, and so there are usually hydroxyl (–OH) groups, strained ring structures, possible double bonds (M=O), and vacant sites.



- There are therefore a variety of possible adsorption sites:
  - metal sites
  - oxygen sites
  - hydroxyl groups
  - vacant sites
- It is more likely that non-competitive adsorption between different species will take place (unlike competitive adsorption on metal surfaces).
- One mechanism that is commonly obeyed is known as the Mars-van Krevelen mechanism.

- In this mechanism, the product molecules contain atoms that were originally in the solid catalyst.
- For instance, oxygen atoms added to a reactant during an oxidation reaction can come from oxygen atoms within the metal oxide structure.
- The catalyst is playing a direct role in the chemical reactions, rather than simply being a substrate.
- The mechanism often involves local changes in metal oxidation state – redox reactions take place.
- The rate law for the Mars-van Krevelen mechanism (no need to prove) is usually of the form

$$r = \frac{k_1 k_2 P_A P_B}{k_1 P_A + k_2 P_B}$$

Example

- Oxidation of CO in air over a copper oxide catalyst at 573 K.
  - Overall reaction: $CO + \frac{1}{2}O_2 \rightarrow CO_2$
- Simplified mechanism:

$$2CuO + CO \rightarrow Cu_2O + CO_2 \qquad \text{slow}$$

$$Cu_2O + \frac{1}{2}O_2 \rightarrow 2CuO \qquad \text{fast}$$

  - The observed rate is first order with respect to CO, and zeroth order in O_2.
- A more detailed mechanism for this reaction involves
  - CO binds initially on a metal adsorption site.
  - Bound CO then reacts with oxygen atoms in the solid to form carbonate species.
  - The carbonate species decomposes to CO_2 and this creates a vacant oxygen site.
  - The vacant oxygen site is then rapidly filled by adsorption of O_2 gas.
- Note that the product CO_2 contains oxygen atoms that started off in the solid metal oxide catalyst.
- Evidence of Mars-van Krevelen mechanism:
  - Isotope labelling experiments
  - A smooth initial surface becomes rough